Negative Viscosity in Magnetic Fluid: Corrected 2D Derivation

Based on Bacri & Perzynski (1995) and Shliomis & Morozov (1994)

1 Key Differences: 3D vs 2D

In 3D, a spherical particle of radius a has volume $V = \frac{4}{3}\pi a^3$ and rotational friction coefficient $\zeta_r = 8\pi\eta a^3$. In 2D, we consider circular disks of radius a with area $A = \pi a^2$ and rotational friction $\zeta_r^{(2D)} = 4\pi\eta a^2$. The 3D Langevin function $L_3(\xi) = \coth\xi - 1/\xi$ is replaced by the 2D Langevin function:

$$L_2(\xi) = \frac{I_1(\xi)}{I_0(\xi)} \tag{1}$$

where I_0, I_1 are modified Bessel functions. The volume fraction ϕ becomes an area fraction. Vorticity Ω in 2D is a pseudoscalar: $\Omega = \frac{1}{2}(\partial_x v_y - \partial_y v_x)$.

2 Torque Balance on a Single Particle

The viscous torque on a 2D disk rotating at angular velocity ω_p in fluid with vorticity Ω:

$$T_{\text{visc}} = -4\pi\eta a^2(\omega_p - \Omega) \tag{2}$$

The torque density (torque per unit area, multiplied by number density n) is:

$$\Gamma_{\text{visc}} = nT_{\text{visc}} = -4\eta\phi(\omega_p - \Omega) \tag{3}$$

since $\phi = n\pi a^2$, so $n \cdot 4\pi\eta a^2 = 4\eta\phi$.

The magnetic torque density in 2D (a pseudoscalar from the cross product $\mathbf{M} \times \mathbf{H}$):

$$\Gamma_{\text{mag}} = M_x H_y - M_y H_x \equiv (\mathbf{M} \times \mathbf{H})_z \tag{4}$$

At torque balance ($\Gamma_{\text{visc}} + \Gamma_{\text{mag}} = 0$):

$$\boxed{4\eta\phi(\omega_p - \Omega) = (\mathbf{M} \times \mathbf{H})_z} \tag{5}$$

Correction note: The original document Eq. (4) is correct in form. The factor 4 (not 6 as in 3D) is the proper 2D coefficient.

3 Antisymmetric Stress and Rotational Viscosity

The antisymmetric part of the stress tensor contributes a body torque:

$$\sigma_a = \frac{1}{2}(\mathbf{M} \times \mathbf{H})_z \tag{6}$$

The total effective shear stress in 2D:

$$\tau_{\text{total}} = 2\eta\dot{\gamma}/2 + \sigma_a = \eta\dot{\gamma} + \frac{1}{2}(\mathbf{M} \times \mathbf{H})_z \tag{7}$$

Using the torque balance $(\mathbf{M} \times \mathbf{H})_z = 4\eta\phi(\omega_p - \Omega)$ and noting $\Omega = \dot{\gamma}/2$ for simple shear:

$$\tau_{\text{total}} = \eta\dot{\gamma} + 2\eta\phi(\omega_p - \Omega) \cdot \frac{\dot{\gamma}}{\dot{\gamma}} \tag{8}$$

This gives the effective viscosity:

$$\eta_{\text{eff}} = \eta + \eta\phi\left(1 - \frac{\omega_p}{\Omega}\right) = \eta + \Delta\eta \tag{9}$$

$$\boxed{\Delta\eta = \eta\phi\left(1 - \frac{\omega_p}{\Omega}\right)} \tag{10}$$

Correction note: The original Eq. (7) is correct. In 3D the prefactor is $\frac{3}{2}\eta\phi$; in 2D it is $\eta\phi$. This arises because $\frac{1}{2} \cdot \frac{\zeta_r^{(2D)} n}{2} = \frac{1}{2} \cdot \frac{4\eta\phi}{2} = \eta\phi$, whereas in 3D: $\frac{1}{2} \cdot \frac{6\eta\phi}{2} = \frac{3}{2}\eta\phi$.

4 Momentum Equation in 2D

The 2D Navier–Stokes equation with ferrofluid body forces:

$$\rho\frac{D\mathbf{v}}{Dt} = -\nabla p + \eta\nabla^2\mathbf{v} + \mu_0(\mathbf{M} \cdot \nabla)\mathbf{H} + \frac{1}{2}\nabla \times [(\mathbf{M} \times \mathbf{H})_z\,\hat{z}] \tag{11}$$

Correction note: The original Eq. (8) writes $(M \cdot \nabla)H + \frac{1}{2}\nabla \times (M \times H)$. In 2D, $\mathbf{M} \times \mathbf{H}$ is a pseudoscalar (the z-component), so $\nabla \times (\cdot)$ must be interpreted as $\hat{z} \times \nabla[(\mathbf{M} \times \mathbf{H})_z]$ acting in the 2D plane. The Kelvin force $\mu_0(\mathbf{M} \cdot \nabla)\mathbf{H}$ retains its form.

5 Magnetization Relaxation Equation

5.1 Shliomis equation in 2D

The relaxation of magnetization toward equilibrium:

$$\frac{d\mathbf{M}}{dt} = \boldsymbol{\omega}_p \times \mathbf{M} - \frac{1}{\tau_B}(\mathbf{M} - \mathbf{M}_0) \tag{12}$$

where $\times$ in 2D means: for a pseudoscalar ω_p and a 2D vector $\mathbf{M} = (M_x, M_y)$:

$$\omega_p \times \mathbf{M} = \omega_p(-M_y, M_x)$$

Substituting $\omega_p = \Omega + \frac{(\mathbf{M}\times\mathbf{H})_z}{4\eta\phi}$ from the torque balance:

$$\boxed{\frac{d\mathbf{M}}{dt} = \Omega \times \mathbf{M} - \frac{1}{\tau_B}(\mathbf{M} - \mathbf{M}_0) - \frac{1}{4\eta\phi}\mathbf{M} \times (\mathbf{M} \times \mathbf{H})} \tag{13}$$

Correction note: The original Eq. (12) is correct in form. The key 2D change is the factor $4\eta\phi$ (not $6\eta\phi$ as in the 3D Bacri paper Eq. (7) which has the Shliomis $1/6\eta\phi$ coefficient). The triple cross product in 2D expands as:

$$\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) = (\mathbf{M} \cdot \mathbf{H})\mathbf{M} - |\mathbf{M}|^2\mathbf{H}$$

which is the BAC-CAB identity (valid in 2D as well).

6 Fokker–Planck Equation in 2D

In 2D, particle orientation is described by a single angle θ. The probability distribution $W(\theta, t)$ satisfies:

$$\frac{\partial W}{\partial t} + \frac{\partial}{\partial \theta}\Big(\dot{\theta}\, W\Big) = \frac{1}{2\tau_B}\frac{\partial^2 W}{\partial \theta^2} \tag{14}$$

The deterministic angular velocity is:

$$\dot{\theta} = \Omega - \frac{\xi}{2\tau_B}\sin(\theta - \theta_H)$$

where θ_H is the field direction angle and $\xi = \mu_0 m H/kT$.

The full Fokker–Planck equation:

$$\boxed{\frac{\partial W}{\partial t} + \frac{\partial}{\partial \theta}(\Omega\, W) = \frac{1}{2\tau_B}\left[\frac{\partial^2 W}{\partial \theta^2} + \frac{\partial}{\partial \theta}(W\, \xi \sin(\theta - \theta_H))\right]} \tag{15}$$

Correction note: The original Eq. (13) writes $\frac{\partial}{\partial\theta}(W\,\xi\cdot\hat{e})$ which is ambiguous. In 2D, with $\hat{e} = (\cos\theta, \sin\theta)$ and field along $\hat{h}$, the potential term is $\xi\sin(\theta - \theta_H)$, which is the torque from the field on the dipole. The diffusion coefficient in 2D is $D_r = \frac{1}{2\tau_B}$ (the factor $\frac{1}{2}$ instead of the 3D value $\frac{1}{2}$—these happen to agree, but for different reasons: in 3D, $D_r = kT/\zeta_r^{(3D)} = kT/(8\pi\eta a^3) = 1/(2\tau_B^{(3D)})$; in 2D, $D_r = kT/\zeta_r^{(2D)} = kT/(4\pi\eta a^2)$ and $\tau_B^{(2D)} = 2\pi\eta a^2/kT$, giving $D_r = 1/(2\tau_B^{(2D)})$).

7 Equilibrium Magnetization in 2D

The equilibrium distribution is $W_0(\theta) \propto \exp(\xi\cos(\theta - \theta_H))$, giving:

$$\boxed{M_0 = \phi M_s L_2(\xi)\,\hat{h}, \qquad L_2(\xi) = \frac{I_1(\xi)}{I_0(\xi)}} \tag{16}$$

Correction note: The original Eq. (14) correctly identifies $L(\xi_e)$ as the 2D Langevin function but writes $M = \phi M_s L(\xi_e)\frac{\xi_e}{\xi_e}$. The notation should be: $\mathbf{M}_0 = \phi M_s L_2(\xi)\,\hat{h}$ where $\hat{h} = \mathbf{H}/H$.

8 Effective Field Method in 2D

Following Martsenyuk, Raikher, and Shliomis, the nonequilibrium magnetization is parametrized as:

$$\mathbf{M} = \phi M_s L_2(\xi_e)\,\frac{\boldsymbol{\xi}_e}{\xi_e} \tag{17}$$

where $\boldsymbol{\xi}_e$ is the "effective field" (in dimensionless units). Substituting into the magnetization equation and projecting:

$$\frac{d}{dt}\left[L_2(\xi_e)\frac{\boldsymbol{\xi}_e}{\xi_e}\right] = \Omega\times L_2(\xi_e)\frac{\boldsymbol{\xi}_e}{\xi_e} - \frac{1}{\tau_B}L_2(\xi_e)\frac{\boldsymbol{\xi}_e-\boldsymbol{\xi}}{\xi_e} - \frac{1}{2\tau_B\xi_e^2}\left(1-\frac{2L_2(\xi_e)}{\xi_e}\right)\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi}) \tag{18}$$

Correction note: The original Eq. (15) has the correct structure. The key 2D modification is the factor inside the last term: $1-\frac{2L_2}{\xi_e}$ rather than the 3D expression $1-L_2/\xi_e-L_2^2$. This comes from the 2D relation:

$$L_2'(\xi) = 1 - \frac{L_2(\xi)}{\xi} - L_2(\xi)^2$$

(using $\frac{d}{d\xi}\frac{I_1}{I_0} = 1-\frac{I_1}{\xi I_0}-\frac{I_1^2}{I_0^2}$), so the coefficient of the last term is:

$$\frac{1}{2\tau_B\xi_e}\cdot\frac{L_2'(\xi_e)}{L_2(\xi_e)/\xi_e} = \frac{1}{2\tau_B}\cdot\frac{\xi_e L_2'(\xi_e)}{L_2(\xi_e)}$$

Actually, let us redo this more carefully. The radial projection of the effective field equation gives:

$$\boxed{\tau_B\frac{d\xi_e}{dt} = -\frac{L_2(\xi_e)/\xi_e}{dL_2/d\xi_e}(\xi_e-\xi\cos\alpha)} \tag{19}$$

where α is the angle between $\boldsymbol{\xi}_e$ and $\boldsymbol{\xi}$. This is the same structure as the original Eq. (16).

9 Perturbation Expansion ($\Omega\tau_B \ll 1$)

9.1 Zeroth order

At zero vorticity, $\boldsymbol{\xi}_e \parallel \boldsymbol{\xi}$ and:

$$\tau_B\frac{d\xi_e}{dt} = -\frac{L_2(\xi_e)/\xi_e}{L_2'(\xi_e)}(\xi_e-\xi) \tag{20}$$

With $\xi = \xi_0\cos\omega t$ (linearly polarized field along $\hat{x}$), this determines $\xi_e^{(0)}(t)$.

9.2 First order

The first-order perturbation in $\Omega\tau_B$:

$$\mathbf{M}^{(1)} = M^{(0)}F(\xi_e)\,\tau_B\,(\Omega\times\hat{h}) \tag{21}$$

where $M^{(0)} = \phi M_s L_2(\xi_e^{(0)})$ and F satisfies:

$$\boxed{\tau_B \frac{dF}{dt} = 1 - \frac{1}{2}\left(\frac{1}{L_2(\xi_e)} - \frac{1}{\xi_e}\right)\xi_e^{(0)} F\cos(\omega t)} \tag{22}$$

Critical correction: The original Eq. (18) has a factor $\frac{1}{2}$ multiplying the term $\left(\frac{1}{L_2} - \frac{1}{\xi_e}\right)$. In 2D, this factor should indeed be $\frac{1}{2}$. Here is the derivation:

The transverse component of the effective field equation gives (after linearization in Ω):

$$\tau_B \frac{d}{dt}(\xi_e F) = \xi_e - \frac{\xi_e^2 F}{2} \cdot \frac{L_2'(\xi_e)}{L_2(\xi_e)/\xi_e} \cdot \frac{\xi_0 \cos\omega t}{\xi_e}$$

Using $\frac{\xi_e L_2'}{L_2} = \frac{\xi_e}{L_2}(1 - L_2/\xi_e - L_2^2) = \frac{\xi_e}{L_2} - 1 - \xi_e L_2$:

After simplification:

$$\tau_B \frac{dF}{dt} + \tau_B F \frac{\dot{\xi}_e}{\xi_e} = 1 - \frac{F\xi_0 \cos\omega t}{2}\left(\frac{1}{L_2(\xi_e)} - \frac{1}{\xi_e}\right)$$

The $\frac{\dot{\xi}_e}{\xi_e}$ term is absorbed by using the zeroth-order equation, and at leading order this reduces to Eq. (22) above.

10 Magnetic Torque and Viscosity Parameter g

The time-averaged magnetic torque:

$$\langle \mathbf{M} \times \mathbf{H} \rangle = -\langle M_y^{(1)} H_x \rangle = -\phi M_s \langle L_2(\xi_e) F\, \tau_B\, \Omega\, \xi_0 \cos\omega t \rangle \cdot \frac{kT}{\mu_0 m} \tag{23}$$

From the torque balance, the mean magnetic torque is:

$$\langle \mathbf{M} \times \mathbf{H} \rangle = -4\eta\phi g\, \Omega \tag{24}$$

where:

$$\boxed{g = \frac{1}{2}\langle \xi_0 L_2(\xi_e^{(0)}) F\cos(\omega t)\rangle_t} \tag{25}$$

Correction note: The original Eq. (20) has $g = \frac{1}{2}\langle \xi_0 L(\xi_e) F\cos\omega t\rangle$, which is correct for 2D. The factor $\frac{1}{2}$ arises from $\frac{\tau_B}{4\eta\phi} \cdot \phi M_s \cdot \frac{kT}{\mu_0 m} \cdot \frac{1}{\tau_B \Omega} = \frac{M_s kT}{4\eta\mu_0 m} = \frac{1}{2}$ using $\tau_B^{(2D)} = 2\pi\eta a^2/kT$ and $\phi = n\pi a^2$.

11 Final Flow Equation

Taking the curl of the momentum equation and time-averaging:

$$\frac{1}{2}\nabla \times \langle \mathbf{M} \times \mathbf{H} \rangle = -2\eta\phi g\, (\nabla \times \boldsymbol{\Omega}) \tag{26}$$

The total viscous force:

$$\boxed{\text{Total viscous term} = -\eta(1 + \phi g)\nabla^2 \mathbf{v} \quad \Longrightarrow \quad \eta_{\text{eff}} = \eta(1 + \phi g)} \tag{27}$$

Therefore:

$$\boxed{\Delta\eta = \eta\phi g} \tag{28}$$

and

$$\boxed{\omega_p = (1-g)\Omega} \tag{29}$$

These final results (original Eqs. 23–24) are correct.

12 Low-Field Limit ($\xi_0 \ll 1$): Analytical Result

For $\xi_0 \ll 1$, $L_2(\xi) \approx \xi/2$ (2D) instead of $\xi/3$ (3D). Then $\xi_e \approx \xi = \xi_0 \cos\omega t$ at zeroth order, and the F equation becomes:

$$\tau_B \frac{dF}{dt} = 1 - F\cos\omega t \cdot \frac{\xi_0}{2} \cdot \left(\frac{2}{\xi_0 \cos\omega t} - \frac{1}{\xi_0 \cos\omega t}\right) = 1 - \frac{F}{2}$$

Wait—this requires more care. For $\xi_0 \ll 1$, $L_2(\xi_e) \approx \xi_e/2$, so $1/L_2 \approx 2/\xi_e$ and $1/\xi_e = 1/\xi_e$, giving $(1/L_2 - 1/\xi_e) = 1/\xi_e$. Then:

$$\tau_B \dot{F} = 1 - \frac{F\xi_0 \cos\omega t}{2\xi_e}$$

At zeroth order $\xi_e \approx \xi_0|\cos\omega t|$, but the effective field tracks the magnitude, so more carefully:

In the weak-field linear regime, we can work directly with the phenomenological equation. The linearized magnetization equation gives $M = \chi H/(1 + i\omega\tau_B)$ in Fourier space, with $\chi = n\mu_0 m^2/(2kT)$ (2D susceptibility, factor 2 not 3).

The rotational viscosity becomes:

$$\boxed{\Delta\eta(\omega) = \frac{1}{4}\eta\phi\xi_0^2 \frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}} \tag{30}$$

Correction note: The original SI-equations document Eq. (14) has $\frac{1}{4}\eta\phi\xi^2\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}$. In 3D this prefactor is $\frac{3}{2}\eta\phi\cdot\frac{\xi^2}{6}\cdot\ldots = \frac{1}{4}\eta\phi\xi^2\ldots$ from the Shliomis result. In 2D: $\Delta\eta = \eta\phi\cdot g$ with $g \sim \frac{\xi_0^2}{4}\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}$. So the formula has the same form. The critical frequency where $\Delta\eta$ changes sign is $\omega_0 = 1/\tau_B$ in both 2D and 3D.

13 High-Frequency Limit

At $\omega\tau_B \gg 1$:

$$\boxed{\Delta\eta \sim -\frac{\phi}{4\eta}\left(\frac{\mu_0 M_s H_0}{\omega}\right)^2 \quad (\omega\tau_B \gg 1)} \tag{31}$$

Correction note: The original SI-equations Eq. (15) has a factor $1/6\eta$ (the 3D result). In 2D, the coefficient changes to $1/4\eta$ because the friction coefficient ratio changes from $6\eta\phi$ to $4\eta\phi$.

14 Summary of Corrections

Quantity	**3D (Bacri)**	**2D (Corrected)**	**Original 2D**
Friction coeff.	$6\eta\phi$	$4\eta\phi$	$4\eta\phi$ ✓
Rotational viscosity	$\frac{3}{2}\eta\phi(1-\omega_p/\Omega)$	$\eta\phi(1-\omega_p/\Omega)$	$\eta\phi(1-\omega_p/\Omega)$ ✓
Langevin function	$\coth\xi - 1/\xi$	$I_1(\xi)/I_0(\xi)$	Stated but not used ✓
σ_a	$\frac{1}{2}(M\times H)$	$\frac{1}{2}(M\times H)_z$	$\frac{1}{2}M\times H$ ✓
FP diffusion	$D_r = 1/(2\tau_B)$	$D_r = 1/(2\tau_B^{(2D)})$	$1/(2\tau_B)$ ✓
High-freq. limit	$-\phi/(6\eta\omega^2)$	$-\phi/(4\eta\omega^2)$	Not given
$\Delta\eta = \eta\phi g$	$\frac{3}{2}\eta\phi g$	$\eta\phi g$	$\eta\phi g$ ✓
Eff. field coeff.	$1 - L_3'/L_3\cdot\xi$	Uses L_2'	Ambiguous

The main errors/issues in the original 2D derivation were:

1. **Eq. (6):** The total shear stress formula $\tau = 2\eta\Omega + 2\eta\phi(\Omega-\omega_p)$ has incorrect prefactors. It should be $\tau = \eta\dot\gamma + 2\eta\phi(\Omega-\omega_p)$ where $\dot\gamma = 2\Omega$.

2. **Eq. (13):** The Fokker–Planck equation's potential term $\frac{\partial}{\partial\theta}(W\,\xi\cdot\hat e)$ should be $\frac{\partial}{\partial\theta}(W\,\xi\sin(\theta-\theta_H))$.

3. **Eq. (14):** The magnetization direction notation ξ_e/ξ_e is unclear; it should be $\hat h_e = \boldsymbol{\xi}_e/\xi_e$.

4. **Eq. (15):** The coefficient $1-2L_2/\xi_e$ in the last term needs to use the proper 2D Langevin derivative $L_2' = 1 - L_2/\xi - L_2^2$, not simply $1-2L/\xi$.

5. **Eq. (19):** The torque formula $\Gamma = -2\eta\phi\Omega[\xi_0 L(\xi_e)F\cos\omega t]$ should not have the factor 2; it should be $\Gamma = -4\eta\phi\Omega\cdot g$ where g contains the $1/2$.

6. **Eq. (21):** Should read $\frac{1}{2}\nabla\times\langle M\times H\rangle = -2\eta\phi g\,\nabla^2\mathbf{v}/(2)$.